

Chapter 6: Renormalization Groups:

Exercise #1:
(pg 321)

(from problem) "One-dimensional q -state
Potts spin model"

$$\beta H = -K \sum_{i=1}^N \delta_{s_i, s_{i+1}} \quad \text{where } s_i = (1, 2, \dots, q)$$

Fixed points:

$q=0$ (unstable): Hamiltonian is zero
with random spins. K
because K -coefficient.

$q=2$ (stable): Hamiltonian has
positive values. At
low-temperature, H
is infinity, same with
 K -coefficient.

$q>2$: $\delta_{s_i, s_{i+1}}$ is not regular
and a ferromagnet.

Exercise #2:
(pg 342)

(pg 337) "Critical exponents"

Symbol	Name	Relationship
α	Temperature	Specific heat $\sim t ^{-\alpha}$
β	Order Parameter	Magnetic field $\sim -t^\beta$
γ	Susceptibility	Susceptibility $\sim t ^{-\gamma}$

δ	Field Dependence	Magnetic field $\sim h ^{1/\delta}$
ν	Correlation length	Root distance to T _{em} $\sim t ^{-\nu}$
η	correlation function	—

Correlation length:

$$\xi \sim |t|^{-\nu}$$

$$\text{Scaled: } b \xi = (b^{y_t} \cdot t)^{-\nu}$$

$$1 = -y_t \cdot \nu$$

$$\nu = 1/y_t$$

Susceptibility:

$$C(r) \sim \begin{cases} r^{2-d-\eta} & r \ll \xi \\ \exp(-r/\xi) & r \gg \xi \end{cases} \dots (\text{pg } 337)$$

$$\text{Scaled: } b^{y_n} C(r) \sim b^{2-d-\eta} \cdot r^{2-d-\eta}$$

$$y_n = 2-d-\eta$$

$$\eta = 2-d-y_n$$

Correlation Function:

(6.3) "Magnetic Susceptibility"

$$\chi = -\partial_H^2 \cdot F|_{H=0}$$

$$= T \partial_H^2 \cdot \ln Z|_{H=0}$$

$$= \beta \int dr dr' \cdot C(r-r')$$

$$\chi = \xi$$

$$\chi = \beta \int d^d r \cdot C(r)$$

$$= \beta \int d^d r \cdot r^{2-d-\eta}$$

... (pg 337)

$$= \xi^{2-\eta}$$

(pg 337) "correlation length"

$$\xi = |t|^{-\nu}$$

(pg 337) "susceptibility"

$$\chi \sim |t|^{-\gamma}$$

Thus,

$$|t|^{-\gamma} \sim (|t|^{-\nu})^{2-\eta}$$

$$\sim |t|^{-\nu(2-\eta)}$$

$$\gamma = \nu(2-\eta)$$

$$= \frac{2y_n - d}{y_t}$$

Order parameter:

$$M \sim (-t)^\beta$$

$$\text{Scaled: } M \sim (-t)^{(d-y_n)/y_t}$$

$$\beta = (d-y_n)/y_t$$

Critical Temperature:

$$C \sim |t|^{-\alpha}$$

$$\text{Scaled: } C \sim |t|^{-(2y_t - d)/y_t}$$

$$\alpha = 2 - d/y_t$$

Field dependence:

$$M \sim |h|^{1/\delta}$$

$$\text{Scaled: } M \sim |h|^{d - y_h / y_H}$$

$$\delta = \frac{y_H}{d - y_H}$$

Exercise #3:

(pg 348)

Parameters:

(6.26) "Action potential- Quadratic"

$$S[\phi] \approx \int d^d r \left(\frac{r}{2} \phi^2 + \frac{1}{2} (\nabla \phi)^2 - h \phi \right)$$

$$\approx \int d(ar)^n \left(\frac{r}{2} \phi^2 + \frac{1}{2} (\nabla \phi)^2 - h \phi \right)$$

$$\approx a^n \int d^n r \left(\frac{r}{2} \phi^2 + \frac{1}{2} (\nabla \phi)^2 - h \phi \right)$$

Propagator lines:

$$n^{\text{th}} \text{ order} - a^{n-4n/2} = a^{-L}$$

$$n-1 \text{ perturbation: } a^{n-4n/2} \circ a^{-1} = a^{-L}$$

$$a^{n-4n/2} = a^{-L+1}$$

Exercise #4:
(pg 357)

Pair contraction:

(from problem) "Matrix field"

$$X \equiv \langle \text{tr}(A_{q,p} \circ W_{p+q} \circ W_p) \text{tr}(A_{q',p'} \circ W_{p'+q} \circ W_{p'}) \rangle$$

where $A, A' \in O(N)$ = matrix fields

$$= \prod_p (\text{tr}(A_{q,p} \circ W_{p+q}) \circ \text{tr}(A_{q',p'} \circ W_{p+q})$$

$$- \text{tr}(A_{q,p} W_{p+q} \circ A_{q',p'} \cdot W_{p+q}))$$

$$= \prod_p (\text{tr}(A_{q,p} \circ W_{p+q}) \circ \text{tr}(A_{q',p'} \cdot W_{p+q})$$

$$- \prod_{p+q} \text{tr}(A_{q,p} \circ W_{p+q}) \text{tr}(A_{q',p'} \cdot W_{p+q})$$

$$- \text{tr}(A_{q,p} \circ A_{q',p'}))$$

$$= \prod_p \prod_{p+q} \text{tr}(A_{q,p} \circ A_{q',p'})$$

Contractions (1-4)(2-3) & (1-3)(2-4):

(1-4) contracts a tensor, $T_{abcd} = T_{abca}$

(2-3) contracts a tensor, $T_{abcd} = T_{abdc}$

(1-3) Contracts a tensor, $T_{abcd} = T_{badc}$

(2-4) contracts a tensor, $T_{abcd} = T_{abcb}$

Exercise #5:

(pg 359)

(from problem) "Critical exponent"

$$\gamma_h = 2 + \epsilon(N-3)/2(N-2) + O(\epsilon^2)$$

(pg 355) "Dimensions expanded"

$$d = d_c + \epsilon$$

(from problem) "Magnetic field perturbation"

$$h \int d^d r \operatorname{tr} [g + g^{-1}] = b^d h \int d^d r \operatorname{tr} [g + g^{-1}]$$

$$= b^{d_c + \epsilon} \cdot h \int d^d r \cdot \operatorname{tr} [g + g^{-1}]$$

$$= b^{2+\epsilon} \cdot h \int d^d r \cdot \operatorname{tr} [g + g^{-1}]$$

$$\approx b^{\gamma_h} \cdot h \int d^d r \cdot \operatorname{tr} [g + g^{-1}]$$

Susceptibility:

(6.22) "Susceptibility"

$$\eta = 2 + d - 2\gamma_h$$

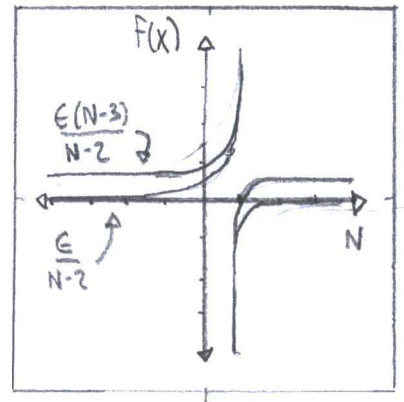
$$= 2 + d - 2 \left(2 + \frac{\epsilon(N-3)}{2(N-2)} \right)$$

$$= \frac{\epsilon(N-3)}{N-2}$$

... at $d=2$

$$N-2$$

$$\approx \frac{\epsilon}{N-2}$$



Exercise #6:

(pg 360)

SU(N) - models:

$$\langle \text{tr}(AW_p) \text{tr}(A'W_{p'}) \rangle = -\frac{\prod_p}{2} \left(\text{tr}(AA') - \frac{1}{N} \text{tr}(AA') \right)$$

(6.34) "SU(N) - completeness relation"

$$\sum_{a=1}^{N^2-1} T_{ij}^a T_{kl}^a = \frac{1}{2} \delta_{il} \delta_{jk} - \frac{1}{2N} \delta_{ij} \delta_{kl}$$

$$P_\mu P_{\mu'} \langle \text{tr}(\Phi_{\mu,q} W_{p+q} W_{-p}) \text{tr}(\Phi_{\mu',q'} W_{p'+q'} W_{-p'}) \rangle_W$$

$$= -P_\mu P_{\mu'} \frac{\prod_{p+q}}{2} \left(\text{tr}(\Phi_{\mu,q} W_{-p}) \text{tr}(\Phi_{\mu,q} W_{-p}) - \frac{1}{N} \text{tr}(\Phi_{\mu,q} W_{-p}) \text{tr}(\Phi_{\mu,q} W_{-p'}) \right)$$

$$= \frac{\prod_p \prod_{p+q}}{4} P_\mu P_{\mu'} \text{tr}(\Phi_{p,-q} \circ \Phi_{\mu',-q})$$

RG Equation:

$$\frac{d \ln \lambda}{d \ln b} = \frac{d}{d \ln b} \left[n \left(\lambda \left(\frac{(N-2)\lambda}{4\pi} - \epsilon \right) \ln b + \lambda \right) \right]$$

$$= \frac{\lambda \left(\frac{(N-2)\lambda}{4\pi} - \epsilon \right)}{b \left(\lambda \left[\frac{(N-2)\lambda}{4\pi} - \epsilon \right] \ln b + \lambda \right)}$$

$$\simeq \frac{N\lambda}{8\pi} - \epsilon \quad \dots \text{When } b > 1$$

Exercise #7:
(pg 364)

(from exercise) "Nonlinear σ -model action"

$$S = \int d^2x (\partial n \cdot \partial n)$$

$$= \int d^2x ((\partial \theta)^2 + \sin^2 \theta (\partial \phi)^2)$$

Topological defect:

$$d\theta = \left(\frac{2r}{1+r^2} \cos \theta, \frac{2r}{1+r^2} \sin \theta \right)$$

$$d\phi = (-\sin \phi, \cos \phi)$$

$$\begin{aligned} (d\theta)^2 &= \left(\frac{4r^2}{(1+r^2)^2} \cos^2 \theta + \frac{4r^2}{(1+r^2)^2} \sin^2 \theta \right) \\ &= \frac{4r^2}{(1+r^2)^2} \end{aligned}$$

$$(d\phi)^2 = (1)$$

$$S = \int d^2x \left(\left(\frac{4r^2}{(1+r^2)^2} + \sin^2 \theta \right) \right) \dots \left(\begin{array}{l} \text{From} \\ \text{exercise} \end{array} \right) \sin \theta = \frac{2r}{1+r^2}$$

$$= \sum_{n=0}^{\infty} \frac{(-g)^n}{n!} \int D\theta e^{\frac{-c}{2} \int d^2r (\nabla\theta)^2} \prod_{n=1}^{\infty} \left(\int dr \frac{e^{i\theta(r)} + e^{-i\theta(r)}}{2} \right)$$

$$= \sum_{N=0}^{\infty} \frac{y_0^{2N}}{(N!)^2} \prod_{i=1}^{2N} \int dr e^{-i \sum_{i=1}^{2N} (-1)^i \theta(r)}$$

Exponential Series

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

... When $y_0 = g/2$
and positive, also
negative phases
appear equally.

Cosine-exponential Relationship:

$$\cos\theta = \frac{e^{i\theta} + e^{-i\theta}}{2}$$

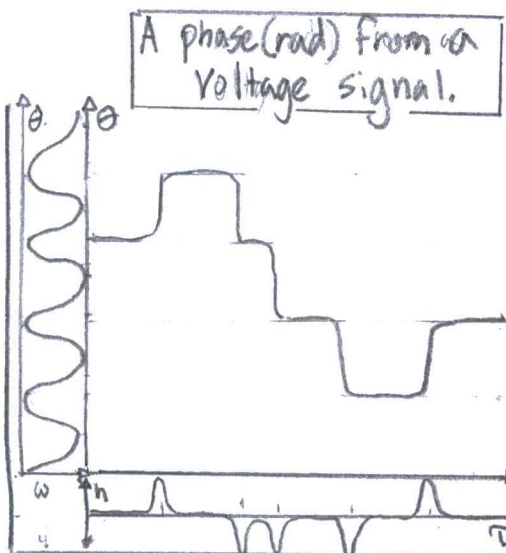
$$\approx \sum_{N=0}^{\infty} \frac{y_0^{2N}}{(N!)^2} \int Dr e^{4\pi^2 J \sum_{i<j} C(r_i, r_j)}$$

... (from problem) "self-interaction"

$$\langle \theta(r) \theta(r') \rangle = C(|r-r'|) / c$$

$$= 8\pi^2 J C(|r-r'|)$$

Problem 6.7.1 - Dissipative Quantum Tunneling - Strong Potential Limit:



Time of an instanton-event

$$= 0 \int d^2x \frac{r^2}{(1+r^2)^2}$$

$$\lim_{r \rightarrow \infty} S = \lim_{r \rightarrow \infty} 0 \int d^2x \frac{r^2}{(1+r^2)^2}$$

$$= 0, \text{ "finite"}$$

Exercise #8: (6.45) "Solutions to two-dimensional Poisson Equation"

(pg 366)

$$\Delta \phi(r) = 2\pi C(r) \quad C(r) = \frac{1}{2\pi} \ln(r)$$

Exercise #9: (6.47) "Sine-Gordon Model"

(pg 367)

$$S[\theta] = \frac{c}{2} \int d^2r (\nabla \theta)^2 + g \int d^2r \cos \theta$$

(pg 322) "Partition function"

$$Z = \int D\theta e^{-S[\theta]}$$

$$= \int D\theta e^{-\frac{c}{2} \int d^2r (\nabla \theta)^2 - g \int d^2r \cos \theta}$$

$$= \int D\theta e^{-\frac{c}{2} \int d^2r (\nabla \theta)^2} \cdot \sum_{n=0}^{\infty} \frac{(-g)^n}{n!} \left(\int d^2r \cos \theta \right)^n$$

Exponential

Series:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$= \int Dq e^{-\frac{1}{4\pi g} \int d\omega \sum_{i,j=0} \epsilon_i(\omega) \epsilon_j(\omega) e^{-i\omega(\tau_i + \tau_j)} \frac{|f(\omega)|^2}{|\omega|}}$$

$$\rightarrow -n \zeta_{inst}$$

$$= \int d\tau \int Dq e^{-\frac{1}{4\pi} \int d\omega \sum_{i,j} q(\omega)^2 \frac{g|\omega|}{4|f(\omega)|^2} \cdot e^{i\omega(\tau_i + \tau_j)} + i \sum \epsilon(\omega) q(\omega)}$$

$$\rightarrow -n \zeta_{inst}$$

$$= \int d\tau \cdot e^{-n \zeta_{inst}} \cdot \int Dq e^{\frac{-g}{4\pi} \int d\omega \sum_{i,j} \frac{q(\omega)^2 |\omega|}{4|f(\omega)|^2} e^{i\omega(\tau_i + \tau_j)} + i \sum \epsilon(\omega) q(\omega)}$$

$$b) Z = \int Dq e^{\frac{-g}{4\pi} \int d\omega \sum_{i,j} \frac{|q(\omega)|^2 |\omega|}{|f(\omega)|^2}} \cdot \int d\tau e^{-n \zeta_{inst} + i \sum \epsilon(\omega) q(\omega)}$$

$$\text{If } \gamma = Z e^{-\zeta_{inst}} \text{ and } T = \int \frac{1}{f(\omega)^2} e^{-i\omega(\tau_i + \tau_j)}$$

then,

$$\begin{aligned} \text{Re}(Z) &= \int Dq e^{\frac{-g}{4\pi T} \sum_{\omega} |q(\omega)|^2 |\omega|} \cdot \gamma \int d\tau \frac{\cos(q(\tau))^n}{n!} \\ &= \int Dq e^{\frac{-g}{4\pi T} \sum_{\omega} |q(\omega)|^2 |\omega|} + \gamma \int d\tau \cos(q(\tau)) \end{aligned}$$

Identities above:

Euler's Formula:	Hubbard-Stratonovich:
$e^{ix} = \cos(x) + i \sin(x)$	$e^{-\frac{a}{2}x^2} = \int dy e^{-\frac{y^2}{2a} + ixy}$

(6.11) "Action of quantum tunneling"

$$S[\theta] = \frac{1}{4\pi g} \int (d\omega) |\omega| |\theta(\omega)|^2 + c \int d\tau \cos(\theta(\tau))$$

(from problem) "Action kinetic energy"

$$S_{\text{kin}}[\theta] = (\ell_m^2/2) \int d\tau \dot{\theta}^2$$

a) IF $\theta_m = -i f_m \omega_m^{-1} \sum_{i=0} e_i e^{i\omega_m \tau}$, then

$$S_{\text{Tot}}[\theta] = S[\theta] + S_{\text{kin}}[\theta]$$

$$= \frac{1}{4\pi g} \int (d\omega) |\omega| \left| -i f_m \omega_m^{-1} \sum_{i=0} e_i e^{i\omega_m \tau} \right|^2$$

$$+ c \int d\tau \cos(\theta(\tau)) + (\ell_m^2/2) \int d\tau \dot{\theta}^2$$

$$= \frac{1}{4\pi g} \int (d\omega) \sum_i e_i(\omega) e_j(\omega) e^{\frac{i\omega_m(\tau_i + \tau_j)}{\omega}} \frac{|f(\omega)|^2}{\omega}$$

$$+ c \quad + n S_{\text{inst}}$$

where $S_{\text{inst}} = \int d\tau (c \cos(\theta(\tau)) + \frac{\ell_m^2}{2} \dot{\theta}^2)$

(3.71) "Field Integral"

$$Z = \int Dq e^{-S[q]}$$

(pg 125) "Free energy"

$$F = -T \ln Z$$

$$= -T \ln \left(\int_{-\infty}^{\infty} Dm e^{-\frac{T}{2} \sum_n \int (dq) (\sigma + q^2 + \frac{|\omega|}{T_2}) |m_q|^2} \right)$$

$$= \dots \left[\int_0^{\infty} Dm e^{-\frac{1}{2} (x + (x+a)^2 m^2)} \approx \frac{1}{(x+a) \sqrt{\pi}} \tanh^{-1} \left(\frac{1}{x+a} \right) \right]$$

$$\approx 2 \cdot L^d \int dq \tanh^{-1} \left(\frac{\omega / T_2}{\sigma + q^2} \right)$$

$$\approx 2 \cdot L^d \int dq \int_0^{T_2} d\omega \coth \left(\frac{\omega \beta}{2} \right) \tanh^{-1} \left(\frac{\omega / T_2}{\sigma + q^2} \right)$$

$\dots T\beta \gg 2$

Derivative:

$$\frac{\partial F}{\partial \sigma} = \frac{\partial}{\partial \sigma} \left[2 \cdot L^d \int dq \int d\omega \coth \left(\frac{\omega \beta}{2} \right) \tanh^{-1} \left(\frac{\omega / T_2}{\sigma + q^2} \right) \right]$$

$$= -2 L^d \int dq \int d\omega \coth \left(\frac{\omega \beta}{2} \right) \frac{\omega / T_2}{(\sigma + q^2)^2 + (\omega / T_2)^2}$$

b) Renormalization (dm) and dynamical exponent (Z):

(from problem) "Rescale"

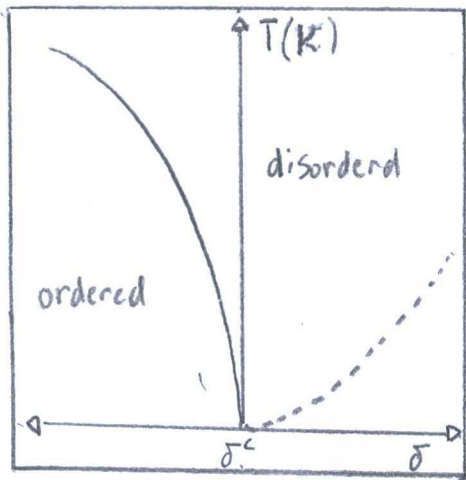
$$q' = Z q, \quad \omega' = b^z \omega, \quad m'_q = b^{-dm} \cdot m_q$$

(from problem) "Action for itinerant magnetism"

Exponential Series:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

Problem 6.7.2 - Quantum Criticality:



Compare and Contrast:

Critical Exponents	Debye / Einstein Model
- Specific heat - Temperature - Order parameter - Spin dependent	- Specific heat - Temperature - Proportional - Substance dependent

↖ closer to constants in 2025.

(pg 375) "Action for itinerant magnetism"

$$S[m] = \frac{T}{2} \int_n \int (dq) \left(\delta + q^2 + \frac{|W_n|}{T_2} \right) |m_2|^2 + \frac{U}{4} \int dx m^4(x)$$

(from problem) "Gaussian theory"

$$\delta > 0, U = 0$$

(3.71) "Field integral"

$$\begin{aligned}
 Z &= \int Dm e^{-S[m]} \\
 &= \int Dm e^{-\frac{T}{2} \sum_n \int (dq) \left(\delta + q^2 + \frac{|W_n|}{T_2} \right) |m_2|^2}
 \end{aligned}$$

$$S[m] = \frac{T}{2} \sum_n \int (dq) (\delta + q^2 + \frac{|w_n|}{T_q}) |m_q|^2 + \frac{u}{4} \int dx m^4(x)$$

$$= \frac{1}{2} \int_n^{N/b} (dq) \int_0^{T_q} (dw) (\delta + q^2 + \frac{|w_n|}{T_q}) |m_q|^2$$

... from statement $u=0$, $T \sum_n = \int (dw)$

$$= \frac{b^{-d}}{2} \int_n^{N/b} (dq)^d \int_0^{T_q} (dw) (\delta + (q^2 b^{-2}) + \frac{|b^{-z} w|}{T_q})$$

$$\times |b^{dm} \cdot m_q|^2$$

If $-d - z - z + 2dm = 0$

then, $dm = \frac{(d + z + z)}{2}$

$$S[m] = \frac{1}{2} \int_n^{N/b} (dq)^{d+z} \int (dw) (\delta + q^2 + \frac{|w|}{T_q}) |m_q|^2$$

$$= \frac{b^{-d-z}}{2} \int_n^{N/b} (dq)^{d+z} \int (dw) (\delta + q^2 b^{-2} + \frac{|w| \cdot b^{-z+2dm}}{T_q}) |b^{dm} m_q|^2$$

If $-d - z - z - z + 2dm = 0$

then $z = 1$ from previous
equality.

Scaling dimension (y_h):

$$S[m] = \frac{b^{-d-2}}{2} \int (dq) \int (dw) \left(\delta \cdot b^2 + q^2 \cdot b^{-2} + \frac{|w| b^{-2}}{\Gamma_q} \right) \times b^{2dm} \cdot |m_2|^2$$

$$\text{If } -d-2+2y_h-2-z+2dm = -d-2-z+2dm$$

$$y_h = 2$$

Quartic Interaction (y_u):

$$\begin{aligned} S[m] &= \frac{T}{2} \sum_n \int (dq) \left(\delta + q^2 + \frac{|w|}{\Gamma_q} \right) |m_2|^2 + \frac{u}{4} \int dx m^4(x) \\ &= \frac{b^{-d-2}}{2} \int (dq) \int (dw) \left(\delta \cdot b^2 + q^2 \cdot b^{-2} + \frac{|w| b^{-2}}{\Gamma_q} \right) b^{2dm} \cdot |m_2|^2 \\ &\quad + \frac{b^{y_u} u}{4} \int dx b^{4dm} \cdot m^4(x) \end{aligned}$$

$$\text{If } -d-2-z+2dm = -d-2-z+2dm + y_u + 4dm$$

$$y_u = -4dm$$

$$= -2(d+3) \quad @ \quad z=1$$

$$c) \langle m_F^2 \rangle = 2 \int_0^1 \int_0^1 (dq) (dw) \coth\left(\frac{w\beta}{2}\right) \frac{\omega \Gamma_q}{(\delta + q^2)^2 + (w/\Gamma_q)^2}$$

Try the integral!

$$\text{If } \delta' = \delta(1 + 2\ln b) + 12uf(\tau)$$

$$u' = b^{4-d-z}$$

$$T' = b^z T$$

$$\text{then } \frac{dT}{d\ln b} = zT$$

$$\frac{du}{d\ln b} = (4-d-z)u$$

$$\frac{d\delta}{d\ln b} = 2\delta + u f(T)$$

$$d) \frac{dT}{d\ln b} = \frac{dT}{du} \dots \text{if } u = \ln b$$

$$= zT$$

$$T(b) = e^{z\ln b + C}$$

$$= C \cdot e^{z\ln b}$$

$$= C \cdot b^z$$

$$= T b^z \dots \text{if } C = T$$

Linear First order
Differential Equation

$$\frac{du}{d\ln b} = (4-d-z)u$$

$$u(b) = ub^{4-d-2} \quad \text{if } c=u$$

Linear First order Nonhomogeneous Differential	Solve	$\frac{d\delta}{d\ln b} = \frac{d\delta}{dx} \quad \text{if } x = \ln b$
	Integrate	$= 2\delta + u f(T)$
	Integrate	$\frac{d\delta}{dx} - 2\delta = u f(T)$
	Factor	$\frac{d\delta}{dx} \cdot e^{-2x} - 2\delta \cdot e^{-2x} = u f(T) \cdot e^{-2x}$
	Simplify	$\frac{d}{dx} (\delta \cdot e^{-2x}) = u f(T) \cdot e^{-2x}$
		$\delta \cdot e^{-2x} = \int u f(T) \cdot e^{-2x} dx$
		$\delta(b) = e^{2x} \left(C + \int u f(T) e^{-2x} dx \right)$
		$= \delta b^2 + b^2 u \int dx \cdot \frac{f(T)}{b^2}$

Problem 6.7.3 - Kondo Effect, Poor mans scaling:

a) (6.60) "Schrödinger equation"

$$\left(\hat{H}_{10} \frac{1}{E - \hat{H}_{00}} \hat{H}_{01} + \hat{H}_{11} + \hat{H}_{12} \frac{1}{E - \hat{H}_{22}} \hat{H}_{21} \right) |\psi_1\rangle = E |\psi_1\rangle$$

(from problem) "sd-Hamiltonian"

$$\hat{H}_{sd} = \sum_{k\sigma} \epsilon_k c_{k\sigma}^\dagger c_{k\sigma} + \sum_{kk'} \left(J_z \hat{S}^z (c_{k\uparrow}^\dagger c_{k'\uparrow} - c_{k\downarrow}^\dagger c_{k'\downarrow}) + J_+ \hat{S}^+ c_{k\downarrow}^\dagger c_{k'\downarrow} + J_- \hat{S}^- c_{k\uparrow}^\dagger c_{k'\uparrow} \right)$$

A + band edge:

$$\left(\hat{H}_{12} \cdot \frac{1}{E - H_{22}} \cdot \hat{H}_{21} \right) | \Psi_1 \rangle = \sum_{kk'} \left(J_z \hat{S}^z (c_{k\uparrow}^\dagger c_{k'\uparrow} - c_{k\downarrow}^\dagger c_{k'\downarrow}) + J_+ \hat{S}^+ c_{k\downarrow}^\dagger c_{k'\downarrow} + J_- \hat{S}^- c_{k\uparrow}^\dagger c_{k'\uparrow} \right)$$

$$\times \left(\frac{1}{E - H_{22}} \right) \times$$

$$\sum_{kk'} \left(J_z \hat{S}^z (c_{k\uparrow}^\dagger c_{k'\uparrow} - c_{k\downarrow}^\dagger c_{k'\downarrow}) + J_+ \hat{S}^+ c_{k\downarrow}^\dagger c_{k'\downarrow} + J_- \hat{S}^- c_{k\uparrow}^\dagger c_{k'\uparrow} \right)$$

$$= J_+ J_- \sum \hat{S}^- c_{k\uparrow} c_{k\downarrow} \frac{1}{E - H_{22}} \hat{S}^+ c_{k\downarrow}^\dagger c_{k\uparrow}^\dagger | \Psi_1 \rangle$$

+ multiple terms ...

$$b) J_+ J_- \sum_{k_s, k_f} \hat{S}^- c_{k\uparrow}^\dagger c_{k\downarrow} \frac{1}{E - H_{22}} \hat{S}^+ c_{k\downarrow}^\dagger c_{k\uparrow}$$

$$= \frac{J_+ J_- \hat{S}^- \hat{S}^+}{E - H_{22}} \sum_{k_s, k_f} \int_{D/b}^{V_0 D} dk_f \cdot c_{k\uparrow}^\dagger \cdot c_{k\downarrow} \cdot c_{k\downarrow}^\dagger \cdot c_{k\uparrow}$$

$$= J_+ J_- \sum_{k_s k_f} V_0 D(1-b') \hat{S}^- \hat{S}^+ c_{k\uparrow}^\dagger c_{k\uparrow} \frac{1}{E - D + \epsilon_k - H_0}$$

... when $H_{zz} = H_0 + D - \epsilon_k$

$$= J_+ J_- \sum_{k_s k_f} V_0 (1-b') \left(\frac{1}{2} - S^z\right) c_{k\uparrow}^\dagger c_{k\uparrow}$$

... When $\hat{S}^- \hat{S}^+ = \frac{1}{2} - S^z$

and $E, H_0, \epsilon \ll 1$